

HAN GAO

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EDUCATION

Master of Statistics, Columbia University, New York, NY

Sep 2025 - May 2026

Bachelor of Data Science and Mathematics, New York University, New York, NY

Sep 2021 - May 2025

Certification: AWS Certified Data Engineer Associate | Option 101

Visa Status: Green Card Candidate

SKILLS

Programming: Python (NumPy, Pandas, Scikit-learn, LightGBM, PyTorch, TensorFlow, Keras), C++, SQL

Modelling: Transformer, NLP (BERT), LLM (Pre-training, Fine-tuning, RLHF), Multi-Agent Systems, Reinforcement Learning

Systems: AWS, Ray, Huggingface, Snowflake, Claude, Jira, Azure, Agile, HPC, Kubernetes, Distributed Training, Jenkins, Zilliz

Course: Gen AI, Machine Learning, Deep Learning, Stats in Finance, Math in Finance, Multi-Asset Portfolio Mgmt

PROFESSIONAL EXPERIENCE

CBRE Investment Management | Data Scientist Intern | Dallas

June 2025 - Aug 2025

- **Full Stack:** Built a GenAI-powered newsletter pipeline (Google Search API + LangChain LLM), leveraging prompt engineering to optimise summarisation and thematic classification. Adopted team-wide, it saved 8+ hours/week and provided timely insights to keep the team ahead of emerging AI trends
- **ML:** Modelled office market dynamics with Bayesian Structural Time Series (CausalImpact) and Transformer, integrating macro/microeconomic indicators and LLM-extracted news themes and NLP-based sentiment analysis
- Designed regime-switching logic and Smooth Decay to detect structural breaks; identified early market risk signals, successfully capturing market turning points up to 2 quarters earlier
- **Research:** Benchmarked multiple LLMs and retrieval techniques to evaluate summarisation accuracy and cost-efficiency, identifying the optimal stack and reducing processing costs by 50%

Vestwell (Unicorn Fintech) | Data Engineer Intern | New York

May 2024 - Aug 2024

- **ML:** Developed Random Forest Classification with 30K data points to identify high-risk transactions and achieved false positive reduction by 20%, with 0.96 R^2 & 4.4% MAPE, resulting in 500 K+ cost savings and reduced review time
- **Data ETL:** Built a PySpark ETL pipeline processing 5M+ structured financial records, integrated into Snowflake + AWS Batch, cutting data processing time by 30% and improving anomaly detection
- **Transformation:** Automated data transformation and QA tests (unit & integration) with Jenkins + Docker, reducing analysts' daily workload by 1 hour
- **Data Integration:** Assisted Analysts in extracting data with SQL; Integrated Snowflake datasets into Sigma for analysis
- Presented insights on fraud risk to senior management with data visualisations in Sigma using non-technical words

ACADEMIC PROJECT

Multi-Factor Equity Strategy - Hypothesis | *Self-Research with Prof Parijat Dube*

July 2025 - Dec 2025

- Designed a multi-factor long-short equity strategy, integrating technical indicators, sentiment, and LLM-derived text factors
- Built an LLM-based RAG + trajectory-aware TSFM pipeline (Hugging Face) that conditions 30-day price trajectories on news/earnings via FiLM-style layers to generate cross-sectional 5-day return alpha signals
- Engineered a Zilliz-based vector memory and multi-head retrieval pipeline to index event/price embeddings, surface similar historical trajectories as auxiliary features, and backtest long-short portfolios under realistic trading frictions (slippage, etc)
- TSFM-RAG improved directional accuracy (Hit Rate = 54.5% vs 44.2%), indicating stronger regime and trend detection

Selected AI Project

Sep 2025 - Nov 2025

- **MCP server for async data-access calls:** Built a CSV/Parquet data-analysis tool and async client enabling Claude Desktop to list, summarize, and query local datasets via natural-language tool calls with robust testing and logging.
- **DPO-based LLM & RAG:** Preference-tuned Qwen3-14B with DPO on YouTube title pairs (LoRA + 4-bit quantization via TRL/Unsloth) and built a semantic search for RAG by sentence-transformer embeddings, Chroma, and MMR retrieval
- **Agentic retrieval tool:** Implemented a ZERO_SHOT_REACT_DESCRIPTION agent with a Streamlit UI for multi-step tool reasoning, retrieval, and document comparison using SerpAPI and custom LangChain tools.

Stock Price Prediction | *PyTorch, LSTM, ARIMAX+ANN, Transformer*

Sep 2024 - Feb 2025

- Built ARIMAX, LSTM, and Transformer with macroeconomic indicators, integrating News sentiment signals (REST API + NLTK) to improve accuracy by 5.1% and executed large-scale training using PyTorch Distributed on HPC (Slurm) clusters.

CBRE Datathon, Forecast Warehouse Completion Rates in Atlanta | *PyTorch, ResNet, CNN*

April 2024